```
import pandas as pd
import numpy as np
from sklearn.linear_model import LogisticRegression
```

```
dia=pd.read_csv('/content/drive/MyDrive/diabeties/diabetcsv (1).csv')
dia.head()
```


{"summary":"{\n \"name\": \"dia\",\n \"rows\": 768,\n \"fields\": [\n   {\n   \"column\": \"preg\",\n   \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 3,\n   \"min\": 0,\n   \"max\": 17,\n   \"num_unique_values\": 17,\n   \"samples\": [\n   6,\n   1,\n   3\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"plas\",\n   \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 31,\n   \"min\": 0,\n   \"max\": 199,\n   \"num_unique_values\": 136,\n   \"samples\": [\n   151,\n   101,\n   112\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"pres\",\n   \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 19,\n   \"min\": 0,\n   \"max\": 122,\n   \"num_unique_values\": 47,\n   \"samples\": [\n   86,\n   46,\n   85\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"skin\",\n   \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 15,\n   \"min\": 0,\n   \"max\": 99,\n   \"num_unique_values\": 51,\n   \"samples\": [\n   7,\n   12,\n   48\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"insu\",\n   \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 115,\n   \"min\": 0,\n   \"max\": 846,\n   \"num_unique_values\": 186,\n   \"samples\": [\n   52,\n   41,\n   183\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"mass\",\n   \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 7.884160320375445,\n   \"min\": 0.0,\n   \"max\": 67.1,\n   \"num_unique_values\": 248,\n   \"samples\": [\n   19.9,\n   31.0,\n   38.1\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"pedi\",\n   \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 0.33132859501277484,\n   \"min\": 0.078,\n   \"max\": 2.42,\n   \"num_unique_values\": 517,\n   \"samples\": [\n   1.731,\n   0.426,\n   0.138\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"age\",\n   \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 11,\n   \"min\": 21,\n   \"max\": 81,\n   \"num_unique_values\": 52,\n   \"samples\": [\n   60,\n   47,\n   72\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\

```
n    },\n    {\n      \"column\": \"class\",\n      \"properties\": {\
n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n
\"samples\": [\n          \"tested_negative\",\n
\"tested_positive\"\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    }\n  ]\
n}","type":"dataframe","variable_name":"dia"}
```


```
ind=dia[['age','mass','insu','plas']]
dep=dia['class']
```

```
lr=LogisticRegression()
```

```
lr.fit(ind,dep)
```

```
LogisticRegression()
```

```
age=int(input("enter the age: "))
mass=int(input("enter the mass: "))
insulin=int(input("enter the insulin level : "))
plasma=int(input("enter the plasama level: "))
pred=lr.predict([[age,mass,insulin,plasma]])
print(pred)
```

```
enter the age: 43
enter the mass: 34
enter the insulin level : 79
enter the plasama level: 123
['tested_negative']
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/base.py:493:
UserWarning: X does not have valid feature names, but
LogisticRegression was fitted with feature names
  warnings.warn(
```